Logistic Regression Classifier Workshop

Predicting Pass/Fail from Study Hours

CSCN8010-26F-Sec1

Foundations of Machine Learning Frameworks

Conestoga College

Group 4

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REPOSITORY

https://github.com/Namypark/LogisticRegressionClassifier_Workshop.git

GROUP 3 REPOSITORY

https://github.com/jcamilochirivi/LogisticRegressionClassifier_Workshop.git

DELIVERABLES

* GitHub repo `LogisticRegressionClassifier_Workshop` with required structure, files, and docs
* Produce one PDF file with the title of the workshop, your names, and a link to your remote GitHub repository.
* Drop the PDF in the box before the due date/time.

SUMMARY

This notebook implements log-loss and logistic regression from scratch in NumPy, predicting exam pass/fail from hours studied for 19 students. Gradient descent reaches weights matching scikit-learns unregularized model, with log-loss 0.2321 (66.5% below baseline), a 2.875-hour decision boundary, and 89.47% accuracy. It also covers learning rates, regularization, and limitations.

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